INTRODUCTION TO NETWORK SYSTEMS · CLASS 3 (LIVE ONLINE · MS TEAMS)

Many trucks, one narrow road

Multiple access, Ethernet, switches, VLANs and Wi-Fi

Class 3 of 5 · the last Layer 2 class

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TWO ACTIVITIES TODAY



The Unmute Game

the Teams audio channel is our shared medium



The Human Switch

one of you is the switch, live in chat

Mic and chat ready — I need volunteers in both hours.

Where we are

Last class of Layer 2

done

Class 1

LIVE L2

Recap of L7–3, link layer, MAC, ARP

done

Class 2

VIDEO L2

Framing and error detection

today

Class 3

ONLINE L2

Multiple access, Ethernet, switches, VLANs, Wi-Fi

next

Class 4

LIVE L1

Transmission media, signals, line coding

Class 5

VIDEO L1

Bandwidth, Shannon, modulation, wrap-up

WARM-UP · 5 MIN · FROM THE CLASS 2 VIDEO

Did you watch it?

Q1 **A PPP sender has the payload byte 7E. What does it transmit?**

- A 7E 5E
- B 7D 5E
- C 5E 7D
- D nothing changes

Q2 **What is the preamble at the start of an Ethernet frame for?**

- A error detection
- B the MAC address
- C clock sync and frame start
- D the VLAN tag

Q3 **A receiver divides the frame by G and gets 011. What does it do?**

- A accept it
- B drop the frame
- C fix the error
- D ask for the CRC again

Q4 **Why does Ethernet use CRC rather than a checksum?**

- A it is simpler
- B it catches burst errors
- C it is required by IP
- D checksums need hardware

By the end of today, you can…

Six learning outcomes for Class 3

1 — Block 1

Explain the three families of multiple access protocols

2 — Block 1

Describe CSMA/CD and binary exponential backoff

3 — Block 1

Explain why an Ethernet frame has a minimum size

4 — Block 2

Fill in a switch table by hand from a sequence of frames

5 — Blocks 2–3

Count collision domains and broadcast domains in a diagram

6 — Block 3

Explain why Wi-Fi avoids collisions instead of detecting them

Outcome 3 looks like trivia. It is the best example in the course of physics shaping a protocol.

Many trucks, one narrow road



- Why sharing a channel is hard
- The Unmute Game: you invent the protocols
- ALOHA, CSMA, CSMA/CD and token passing
- Why Ethernet frames have a minimum size

One channel, many senders

The whole problem of Block 1, in one room

THIS MEETING



If I ask a question and seventy people unmute at once, I hear nothing at all.

One channel. Many senders. No referee.

THE SAME PROBLEM IN NETWORKS

The Wi-Fi wherever you are sitting
every device shares one radio channel

Old Ethernet on one cable
every machine tapped onto the same wire

Satellite and cellular uplinks
thousands of senders, one channel up

Think for a moment: in this call, what happens when two people talk at once — and what do we actually do about it?

Every answer you give will have a protocol name by the end of this block.

What a collision actually is

Not a bump — an addition

Station A sends	`1 0 1 1 0 0 1 0`
Station B sends	`0 1 1 0 1 0 1 1`
The receiver hears	***both signals added together***

The sum matches neither frame. Both are lost.

How is it noticed?

The CRC from the Class 2 video fails, so the frame is dropped. A collision just looks like heavy damage.

Where does it happen?

At the receiver, not at the sender. Remember this — it is what breaks Wi-Fi in Block 3.

And the whole time both frames were in the air, the channel was busy for nothing.

What would a good sharing rule look like?

Agree the wish list before meeting any protocol



One sender gets everything

When only one node has data, it should get the full rate of the channel.



Many senders share fairly

Over time each active node should get roughly an equal share.



No master, no shared clock

Decentralised: no single node whose failure stops everyone.



Simple enough for hardware

This runs on every network card ever made. It cannot be clever.

Spoiler: no protocol wins on all four. Every one we meet today trades one of these away for another.

Only three ideas exist

Every real protocol is one of these, or a mixture



Partition the channel

Cut it into pieces and give one piece to each node.

No collisions, but idle pieces are wasted.



Let everyone try

Send when you like, allow collisions, recover afterwards.

Fast when quiet, collapses when busy.



Take turns

Pass a permission around; only the holder may send.

Fair and predictable, but needs coordination.



Before I explain any of them, you are going to invent all three yourselves.

Six volunteers on microphones. Everyone else scores in chat.

ACTIVITY 1 · 20 MIN · SIX VOLUNTEERS ON MIC

The Unmute Game



The goal

Each of the six volunteers must say one complete sentence to the meeting. Any sentence at all.

What counts as heard

A sentence counts only if no other voice overlapped it. If I could not understand it, it did not arrive.

What counts as a collision

Two voices at once: BOTH sentences are lost and both must be said again later.

Everyone else scores

Type C in chat every time you hear a collision. I count the Cs and the clock at the end of each round.

Five rounds, about 60 seconds each. Volunteers stay unmuted the whole time; everyone else scores in chat.

You need

- six volunteers with working microphones
- everyone else: the chat panel open
- one sentence ready to say — anything
- the raise-hand button (round 5)

Five rounds. I change one rule each time. Do not think about networks yet — just play.

The five rounds

One rule changes each time. Score every round.

1	**Free for all**	Speak whenever you want. No rules at all.
2	**Only on the beat**	I say NOW every five seconds. You may only START speaking straight after a NOW.
3	**Listen first**	If you can hear anyone speaking, wait. Start only when it sounds quiet.
4	**Listen, stop, wait**	As round 3, but the moment you hear another voice, STOP and mute. Wait a random 1–6 seconds, then try again.
5	**Only the raised hand**	Raise your hand in Teams. Only the person I call on may speak.

Round 3 will still produce collisions — and online you can hear exactly why. We come back to it.

Scoreboard

Rd	Time	Coll.
1		
2		
3		
4		
5		

Which round felt fastest?
Which felt fairest?

You just invented every protocol in the chapter

Same five rounds, with their real names

1	Free for all	→	**Pure ALOHA**	send whenever you have something
2	Only on the beat	→	**Slotted ALOHA**	transmissions begin only at slot boundaries
3	Listen first	→	**CSMA**	carrier sense: listen before you transmit
4	Listen, stop, wait	→	**CSMA/CD**	collision detection plus random backoff — classic Ethernet
5	Only the raised hand	→	**Polling**	one master invites each node to speak in turn

Variant: if the speaker names who goes next instead of me, that is token passing. Nobody invented these to be clever.

Partitioning 1: time division

Cut time into slots and give one to each node

ALL SIX NODES ARE BUSY



every slot carries data — no collisions, perfectly fair

ONLY NODE 1 HAS ANYTHING TO SEND



five slots out of six are empty, and node 1 still waits through them

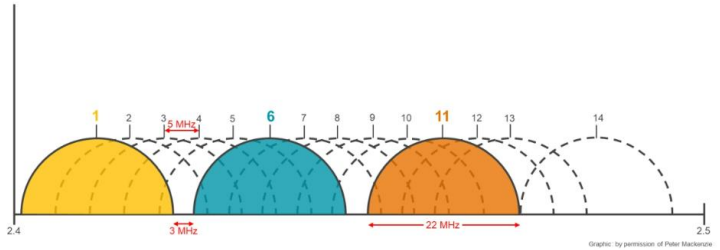
Verdict

- no collisions
- guaranteed share
- idle slots wasted
- one sender gets only 1/N

Breaks wish number 1: a single active node should get the whole channel, and here it never can. Used in telephone trunks and cellular voice slots (2G TDMA).

Partitioning 2: frequency division

Same idea, but split the frequency band instead of time





You already live with this

Wi-Fi channels 1, 6 and 11 on 2.4 GHz are chosen because they do not overlap.

Two access points on channel 1 share it and collide. On 1 and 6 they do not interfere at all.

Same trade as TDM: no collisions and a guaranteed share, but an idle band is simply wasted. A third variant, code division, lets everyone transmit at once using different codes.

https://www.dclessons.com/wifi-frequency-bands

Random access 1: pure ALOHA

ALOHAnet, Hawaii, 1970 — the simplest rule that can work

THE ENTIRE PROTOCOL

When you have a frame, send it.



Anyone who starts in the frame before ours, or during it, destroys it.

18%

maximum efficiency — 1/(2e)

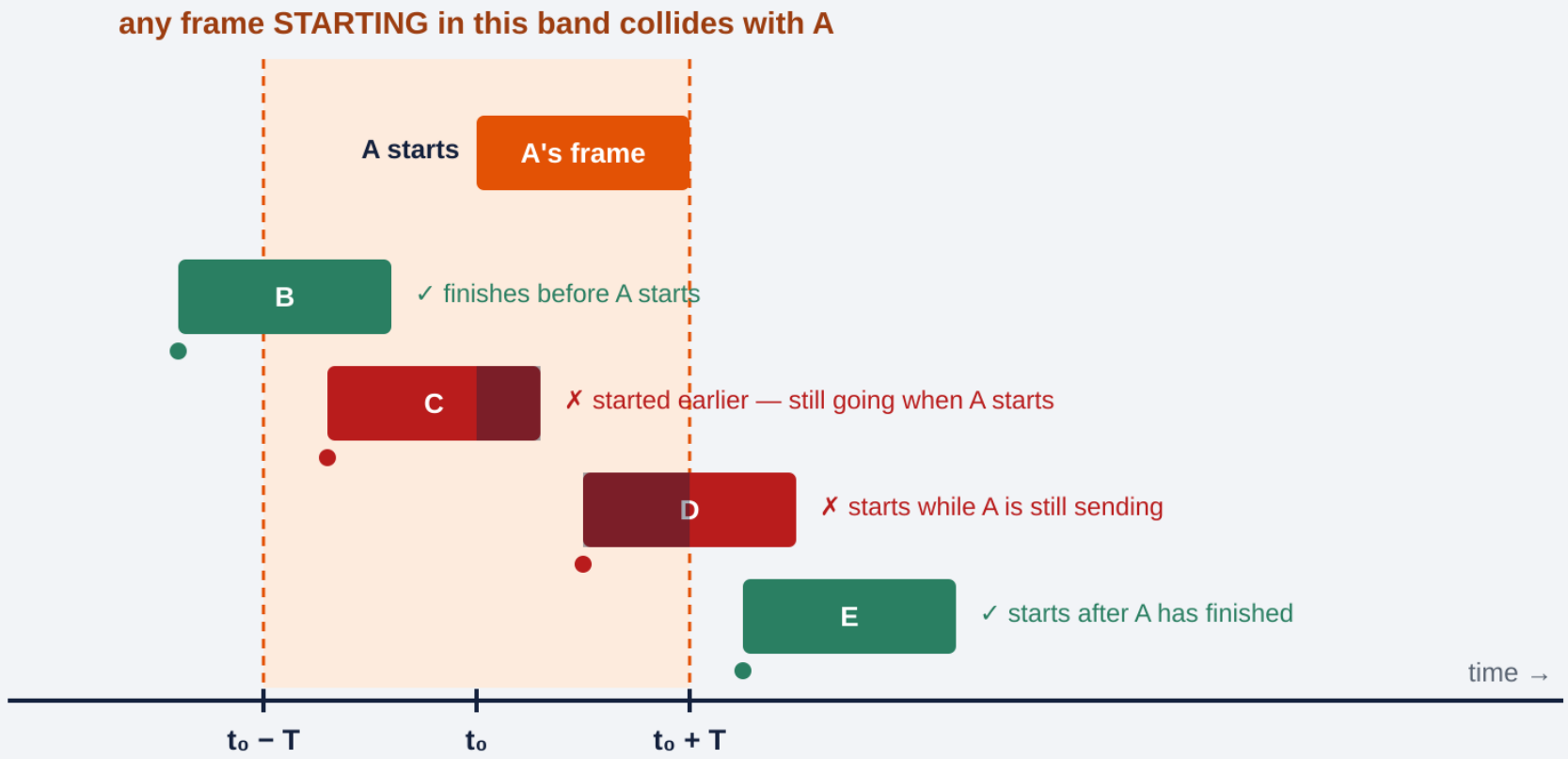
At best, 82% of the channel is wasted on collisions and idle time.

On a collision: no acknowledgment comes back, so retransmit after a RANDOM delay — retrying immediately would collide forever.

So why use it? No coordination whatsoever, and at very low load you send immediately with zero waiting.

The vulnerable period

Node A sends one frame. Which other frames destroy it?



The window

A frame of length T is at risk from any frame that starts in a window of width

2T

The four cases

- B and E miss entirely
- C overlaps the START of A
- D overlaps the END of A
- any overlap at all destroys BOTH frames

Note that C and D are destroyed too — a collision is never one-sided.

Random access 2: slotted ALOHA

One small change doubles the efficiency — that was round 2



Transmissions may only BEGIN at a slot boundary, so a frame can only be hit by one that starts in the same slot.

	Pure ALOHA	Slotted ALOHA
Vulnerable period	2 frame times	**1 frame time**
Maximum efficiency	18% · 1/(2e)	**37% · 1/e**

The price

Everyone now needs a shared clock. In a call I can say NOW; across a network that is real engineering work.

A little coordination bought a doubling in performance. That pattern repeats all through networking.

Still: even at its best, nearly two thirds of the channel is lost. We can do much better by listening first.

Random access 3: listen before you talk

Carrier sense multiple access — that was round 3



Sense the channel

Before transmitting, listen to see whether anyone else is sending.



Idle? Transmit

Send the whole frame.



Busy? Defer

Wait, and try again when it goes quiet.



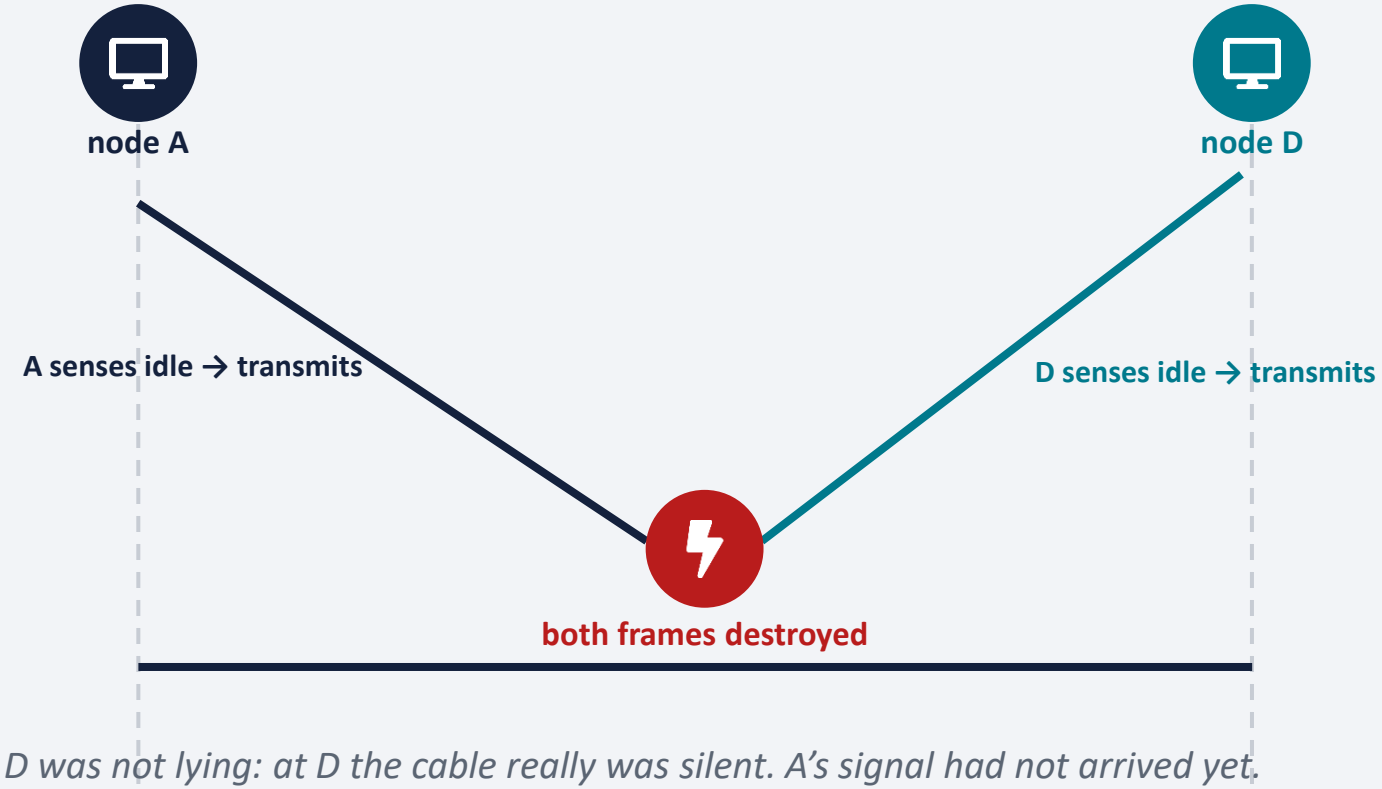
But your groups still collided in round 3. Why?

Two people both heard silence at the same moment, and both were telling the truth.

On a network the same thing happens, and the reason has a name: propagation delay.

Why listening is not enough

Time runs downwards; the cable runs left to right



D was not lying: at D the cable really was silent. A's signal had not arrived yet.

Distance makes sharing hard

The longer the cable, the wider the window in which two nodes can both be wrong.

So Ethernet limits length

A maximum cable length is not a convenience — it is what makes the protocol work.

Detect it and stop: CSMA/CD

That was round 4 — and it is classic Ethernet



1

Keep listening while sending

The node compares what it hears with what it is sending.



2

Different? A collision

Some other signal is on the cable at the same time.



3

Abort immediately

Stop transmitting and send a short jam signal so everyone notices.



4

Wait a random time

Then try again — and the waiting rule is the clever part.

Why abort rather than finish? The frame is already lost. Finishing a doomed 1500-byte frame keeps the channel busy for nothing — on a loaded cable that is the difference between working and collapsing.

Note carefully: this needs a node to transmit and listen at the same time on the same medium. On a cable that is easy. In Block 3 it is exactly what breaks.

Binary exponential backoff

How long should you wait? The protocol works it out from its own failures.

After…	Pick K from	Wait at 10 Mbps
1st collision	0 – 1	0 – 51 µs
2nd collision	0 – 3	0 – 154 µs
3rd collision	0 – 7	0 – 358 µs
4th collision	0 – 15	0 – 768 µs
… up to 10th	0 – 1023	tens of ms
16th attempt	—	**give up, report failure**

Wait = K × 512 bit times. At 10 Mbps, 512 bit times = 51.2 µs.

Why doubling?

A repeated collision is evidence that many nodes are competing. The protocol estimates the load from its own failures — with no measurement and no communication at all.

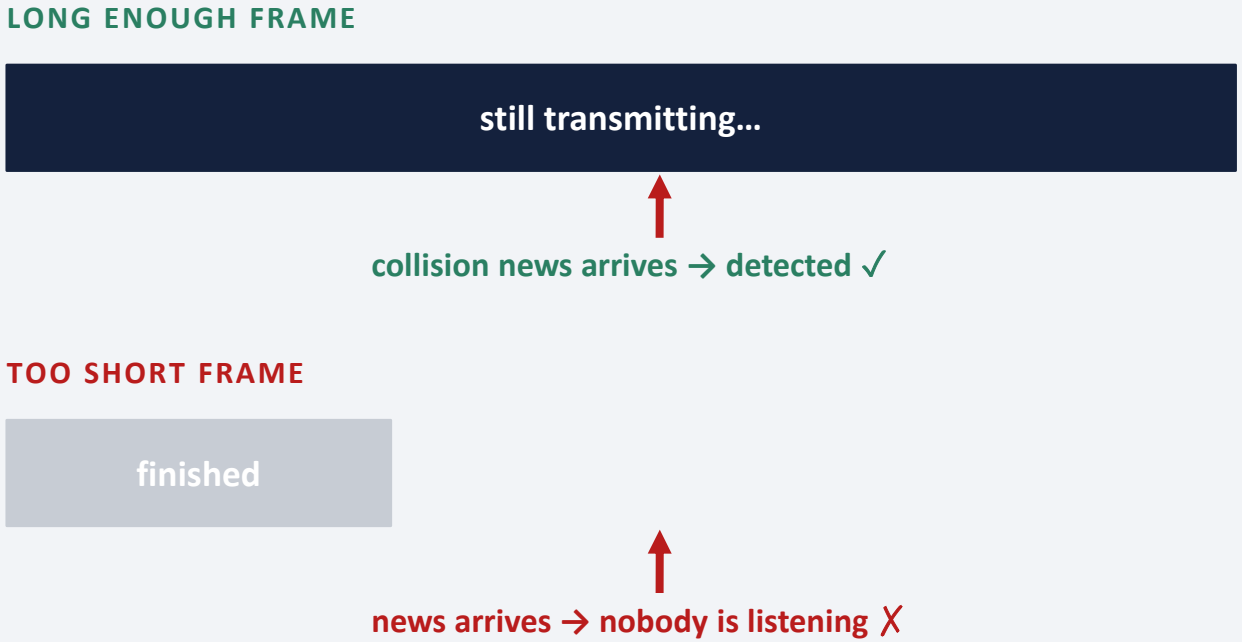
Is it fair?

Not perfectly. A node that just succeeded resets its window and can beat one that has already backed off. This is the capture effect.

Random, not fixed: if both waited the same time they would collide again, forever.

Why an Ethernet frame is at least 64 bytes

The one slide that looks like trivia and is not



The arithmetic

worst-case round trip	`51.2 µs`
at 10 Mbps that is	`512 bits`
which is exactly	`64 bytes`

Shorter frames are padded up to 64 bytes.

The minimum frame size is the speed of light, the maximum cable length and the data rate, written into the frame format. At 100 Mbps the same 512 bits take only 5.12 µs, so the maximum network diameter shrinks tenfold.

The third family: taking turns

Round 5 — only the pen may speak



Polling

One master asks each node in turn whether it has anything to send.

- collision-free and simple
- the master is a single point of failure
- polling itself costs time

Used by Bluetooth.



Token passing

A small special frame circulates. Only the holder may transmit.

- no master, no collisions
- guaranteed maximum waiting time
- if the token is lost, everything stops

Token Ring, and still used in factory networks.

The three families side by side

None of them wins everywhere

Family	Examples	When it shines	When it fails	Collisions?
Partitioning	TDM, FDM	Heavy, steady load from a known set of nodes	Idle nodes still hold their share	**never**
Random access	ALOHA, CSMA, CSMA/CD	Light load, bursty traffic, unknown nodes	Efficiency collapses as load rises	**yes, by design**
Taking turns	Polling, token passing	Predictable delay matters, e.g. factories	Coordination cost; the token can be lost	**never**

Modern Wi-Fi is a mixture: random access to win the first chance to speak, then a reservation for a period of time. We meet it in Block 3.

Exam tip: name the family, then give one strength and one weakness. That is a standard question.

QUICK CHECK · 4 MIN

Before the break

Q1 The maximum efficiency of slotted ALOHA is about…

- A 18%
- B 37%
- C 50%
- D 100%

Q2 Why do collisions still happen with carrier sensing?

- A cables are noisy
- B the CRC is weak
- C propagation delay
- D too many nodes

Q3 Why must an Ethernet frame be at least 64 bytes?

- A to fit the header
- B so the sender is still sending when the collision returns
- C to pad the CRC
- D for the preamble

Q4 In binary exponential backoff, after each collision…

- A the waiting range doubles
- B the frame gets smaller
- C the rate halves
- D the node gives up

BEFORE THE BREAK

Everything in Block 1 assumed one shared cable.

But your laptop is plugged into a switch, not a shared wire. So…

when did your Ethernet card last have a collision?



10-minute break

The answer surprises almost everyone.

Ethernet and switches



- Why the protocol that won is the one you never see working
- Activity 2 — The Human Switch
- Collision domains, broadcast domains, switch vs router

When did your Ethernet card last have a collision?

Be honest — think about your own laptop

Almost certainly never, in your entire life. On a switched, full-duplex network, collisions do not occur. CSMA/CD is still in the standard — and it is switched off.

Wi-Fi still lives there

Everything in Block 1 is exactly how your phone competes for the air right now. The wire escaped; the radio did not.

It explains the frame

The 64-byte minimum, the padding, the preamble — all of it is collision detection, frozen into the header.

Physics shaped a protocol

The clearest example in this course of engineering forced by the speed of light. That idea reappears everywhere.

SAME FRAME FORMAT SINCE 1973

10 Mbps	100 Mbps	1 Gbps	10 Gbps	100 Gbps	400 Gbps
1983	1995	1999	2002	2010	2017

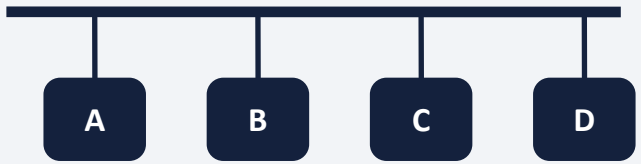
Speed changed by a factor of forty thousand. The header did not change at all. That is what a good design buys you.

The shape changed. The frame did not.

Three generations of the same network

Bus

1973 – early 1990s



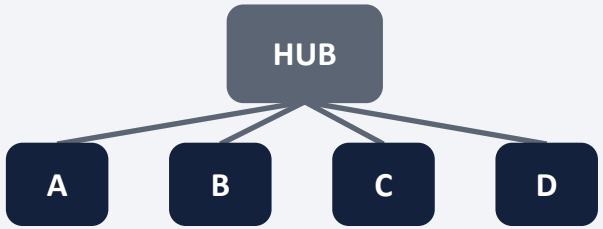
one coaxial cable, every machine tapped on

1 collision domain

A break anywhere kills the whole network.

Star with a hub

early 1990s



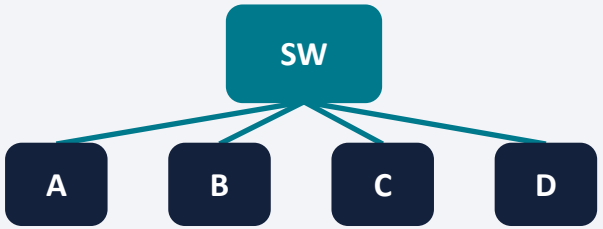
nicer wiring — electrically still one bus

still 1 collision domain

A hub repeats every bit out of every other port.

Star with a switch

mid 1990s onward



each port is its own segment

4 collision domains

Full duplex: send and receive at the same time.

Question before we go on: if two machines transmit at the same moment through a hub, do they collide? — Yes. A hub gives you the tidy wiring of a star and none of the benefit.

The Ethernet frame, properly this time

You met it briefly in the Class 2 video — now every field has a reason



← *framing (Class 2 video)* → ← *addressing (Class 1)* → *error detection* →

Type — one link, many protocols

`0x0800`	IPv4
`0x86DD`	IPv6
`0x0806`	ARP

This two-byte field is why a single cable can carry several unrelated protocols at once.

Why is the payload floor 46 bytes?

`6 + 6 + 2 + 4 = 18`
bytes of header and trailer
`46 + 18 = 64`

The padding floor and the 64-byte minimum are the same constraint, seen from two ends.

The 1500-byte ceiling is the MTU — the number that will reappear when Layer 3 has to fragment.

One family, many wires

Read the naming scheme — do not memorise the rows

1000 megabits per second · **BASE** baseband — the whole wire carries one signal · **T** the medium: T = twisted pair, S / L = short / long wavelength fibre

Standard	Speed	Medium	Max length	Where you meet it
`10BASE-T`	**10 Mbps**	Cat 3 twisted pair	100 m	history — the bus era ending
`100BASE-TX`	**100 Mbps**	Cat 5 twisted pair	100 m	older office equipment
`1000BASE-T`	**1 Gbps**	Cat 5e / Cat 6	100 m	*what is in the wall of almost every campus room*
`10GBASE-T`	**10 Gbps**	Cat 6a	55 – 100 m	servers and campus uplinks
`1000BASE-LX`	**1 Gbps**	single-mode fibre	5 km	building to building
`10GBASE-SR`	**10 Gbps**	multi-mode fibre	300 m	inside a datacentre rack

Every single row uses the frame on the previous slide. That is the whole point of the table.

Hub versus switch

The single most important comparison in this block

Hub

a repeater with sockets

Looks at	nothing at all
Does	copies every incoming bit to every other port
Collision domains	one, shared by everybody
CSMA/CD	mandatory — this is why it exists
Capacity	one conversation at a time
Privacy	everyone can read everyone's traffic

Switch

reads the destination MAC

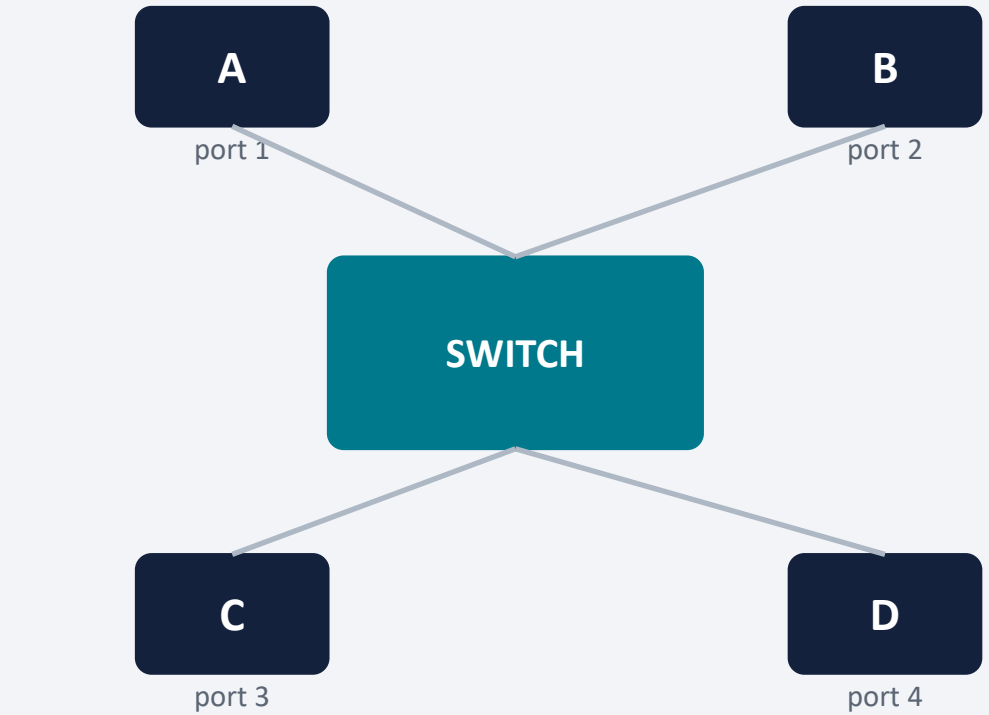
Looks at	the destination MAC address
Does	sends the frame out of one port only
Collision domains	one per port
CSMA/CD	not needed — full duplex, no contention
Capacity	many conversations at full speed at once
Privacy	you normally see only your own traffic

That is the answer to the question I left you with: the switch ended collisions. Now — how does it know which port to use? Nobody configures it. You are about to work it out.

ACTIVITY 2 · RUN IT

Six frames, in this order

Chat predicts first, then the switch answers on mic. I fill in the table.



1	`A → B`	Learns A. B unknown → floods to 2, 3, 4.
2	`B → A`	Learns B. A known → port 1 only. C and D get nothing.
3	`C → A`	Learns C → port 1 only.
4	`A → B`	Both known → one port. The table is doing real work.
5	`D → A`	Learns D. The table is now complete.
6	`A → FF:FF:FF:FF:FF:FF`	Broadcast — flooded anyway. Ask first.

Watch for the two classic mistakes: forwarding before learning, and sending a frame back out of the port it arrived on. Let the chat catch them.

Everything a switch does, in three rules

Write these down — there is nothing else

Learn

RULE 1

Every arriving frame teaches the switch where its sender is. Source MAC + incoming port → the table. The switch never asks; it just pays attention.

Filter and forward

RULE 2

Destination in the table → send it out of that one port and nowhere else. Destination on the port it arrived from → drop it; they are on the same segment already.

Flood

RULE 3

Destination unknown, broadcast or multicast → copy it to every port except the one it came from. This is the expensive case, and it is unavoidable.

Notice what is missing: no configuration, no addresses to assign, no protocol between switches. Plug it in and it works — that is plug and play, and it is most of the reason switches took over. Entries age out after about five minutes, so a device that moves is found again.

What the table looked like

The same six frames, written down

#	Frame	Table after learning	Lookup	What the switch does
1	**A → B**	A : 1	B unknown	**FLOOD to 2, 3, 4**
2	**B → A**	A : 1 B : 2	A known	**forward to port 1**
3	**C → A**	A : 1 B : 2 C : 3	A known	**forward to port 1**
4	**A → B**	A : 1 B : 2 C : 3	B known	**forward to port 2**
5	**D → A**	A : 1 B : 2 C : 3 D : 4	A known	**forward to port 1**
6	**A → broadcast**	A : 1 B : 2 C : 3 D : 4	always flooded	**FLOOD to 2, 3, 4**

The table fills itself from ordinary traffic. Nobody typed anything in. And the first frame to a device that has never spoken is always flooded — once it replies, everyone knows.

What did you notice?

Collect answers first, then show these

It configured itself

The table was built entirely out of normal traffic. No administrator, no protocol, no setup.

Not knowing is expensive

An unknown destination costs everyone: the frame goes to every port, and three machines process something that was not for them.

Conversations run in parallel

A can talk to B while C talks to D, at full speed, at the same time. With a hub, one conversation at a time. The switch multiplied the capacity of the network without changing a single frame.

Broadcast always goes everywhere

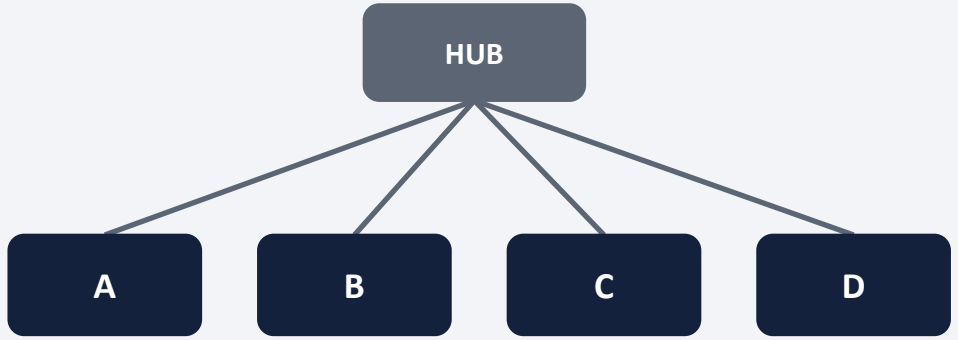
However complete the table is, FF:FF:FF:FF:FF:FF reaches every port. The table cannot help.

Hold on to the fourth one. It is a problem waiting to happen, and after the break we watch it happen.

Collision domains and broadcast domains

Exams love these two, and students mix them up

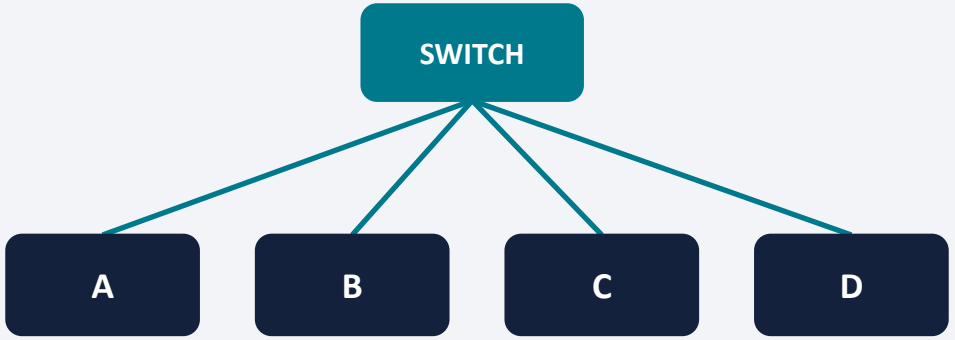
Four PCs on a hub



1 collision domain

1 broadcast domain

Four PCs on a switch



4 collision domains

1 broadcast domain

Now add a router between two switches: still one collision domain per port, but now **two** broadcast domains — because a router does not forward broadcasts.

Switches split collision domains. Routers split broadcast domains. Put that one line in your notes and leave it there.

Switch or router?

Everything from Class 1, now side by side

	SWITCH	ROUTER
Reads up to	Layer 2 — the frame	Layer 3 — the packet inside it
Address it uses	MAC — flat, 48-bit, burned in	IP — hierarchical, assigned
How its table is built	by itself, from ordinary traffic	by configuration or a routing protocol
Scope	within one LAN	between different networks
Broadcasts	floods them everywhere	stops them dead
In the parcel story	the dock worker reading the trip sheet	the sorting hub reading the address label

And if someone sells you a "Layer 3 switch": that is a router with a lot of ports and hardware forwarding. The name is marketing, not a fourth kind of device.

BEFORE THE BREAK

Plug two switches together with two cables instead of one.

Sensible, surely — a spare cable in case one fails.

It takes the whole network down in about four seconds.



10-minute break

It has happened on real university networks. Including ones like ours.

What breaks, and what the air changes

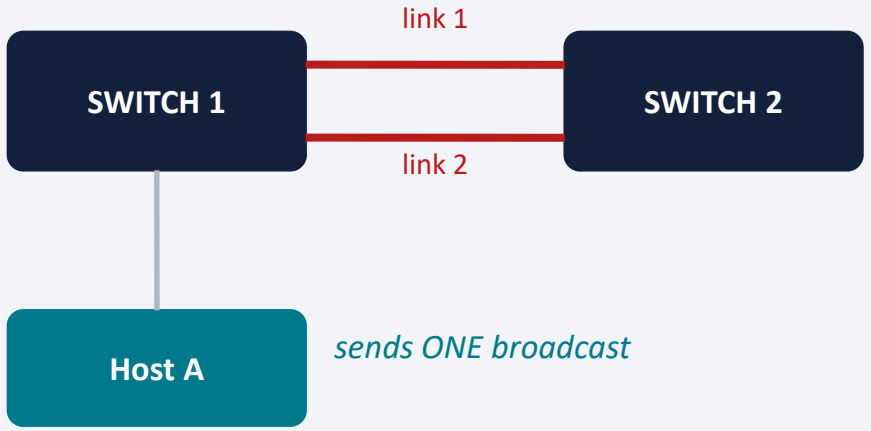
- Loops, broadcast storms and spanning tree
- VLANs — splitting a network in software
- Wi-Fi: why none of Block 1 survives contact with the air



One extra cable, and the network dies

Somebody adds a second link between two switches — for redundancy

Two switches, two cables



copies in flight:	1	2	4	8	16	…

1 — Switch 1 floods it

Destination is broadcast, so it goes out of BOTH links to switch 2.

2 — Switch 2 gets two copies

It floods each one — including back over the other link.

3 — Switch 1 gets them back

And floods them again. Nothing tells either switch it has seen this frame before.

4 — Within seconds

Links saturated, switch CPUs at 100%, MAC tables thrashing because the same source keeps arriving on different ports.

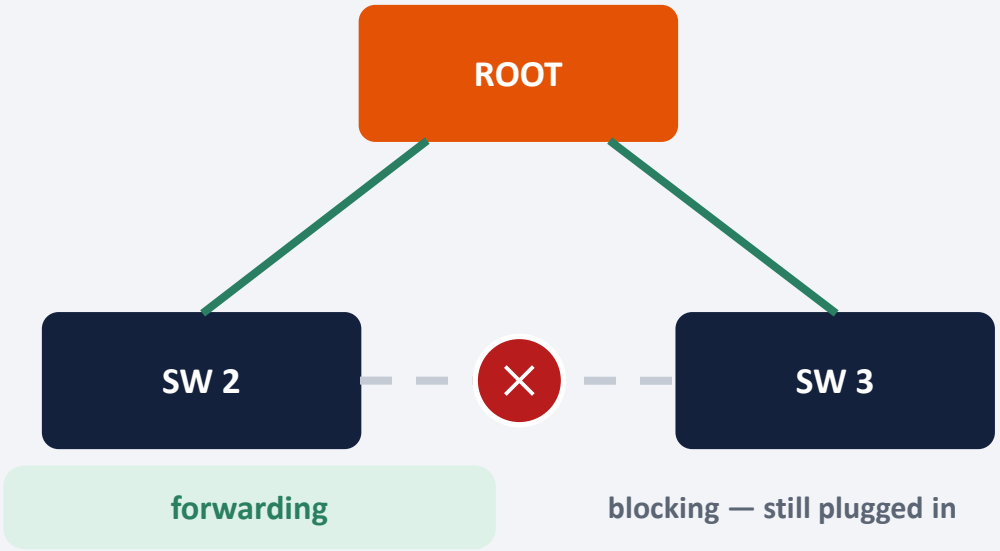
Why does the frame never die? Because an Ethernet frame has no hop count. Layer 3 has TTL; Layer 2 has nothing at all. The frame does not decay — it multiplies. This is a **broadcast storm**, and one cable in the wrong place causes it.

So why is a loop attractive in the first place? Redundancy — if one link fails you want the other. The problem is having both active at once.

Spanning Tree Protocol

Keep the spare cable. Just do not use it yet.

Three switches, three links, one blocked



Elect a root

The switches talk to each other and agree on one of them as the root of the tree.

Find the cheapest way home

Each switch works out which of its links leads towards the root most cheaply.

Block the rest

Any link not needed to reach the root goes into blocking state: physically connected, carrying no data frames.

Keep the redundancy

If an active link fails, a blocked link is unblocked and takes over. You get the spare cable without the loop.

Could we just tell people not to create loops? People make mistakes, and sometimes the loop is a mis-patched cable in a closet nobody can find. The protocol has to assume it will happen.

Classic STP took 30–50 seconds to converge — which is why Rapid STP exists. That is as far as we go today.

Broadcasts go everywhere. That is the problem.

Back to the fourth thing you noticed in the Switch

The problem

Thousands of devices in one broadcast domain. Every ARP request from every machine reaches every other machine.

- wasted work on every device
- anyone can see the broadcasts
- anyone can answer them — remember ARP spoofing

The obvious fix

Buy more routers and split the campus into separate physical networks.

- it works
- it is expensive
- moving one person means re-cabling

The VLAN fix

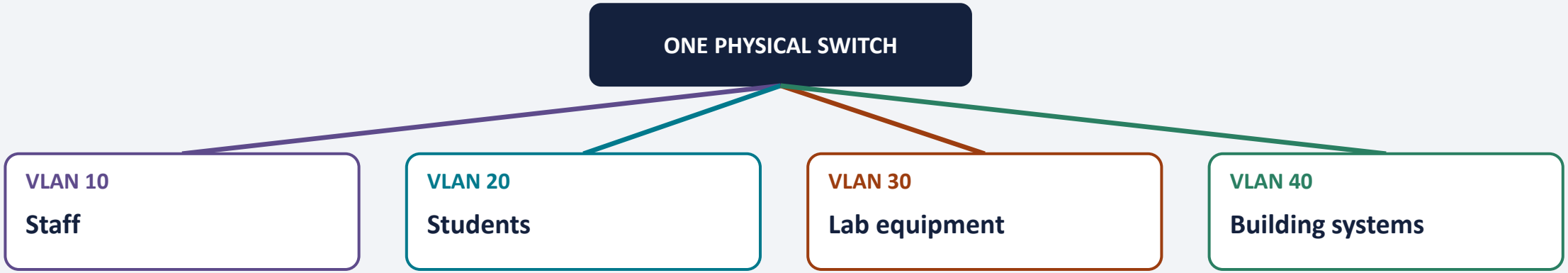
Divide one physical switch into several virtual switches, in software.

- ports 1–8 = VLAN 10
- ports 9–16 = VLAN 20
- a VLAN 10 frame never leaves a VLAN 20 port — not even a broadcast

The consequence worth stating plainly: devices in different VLANs cannot reach each other at Layer 2 at all. To talk, they must go through a router — which is exactly the control we wanted.

This is how the network you are on right now is built

Same switches, same cables in the ceiling — four separate networks



Containment

A student laptop with a virus cannot reach the air-conditioning controllers. The frames simply cannot leave the student VLAN.

Blast radius

A broadcast storm in one teaching lab does not touch the rest of the campus.

Different rules per group

Each VLAN is its own IP subnet, so the firewall can treat staff, students and equipment differently.

Moving people is free

Changing someone's network is a configuration change, not a cabling change.

I sit on the committee that runs this network, so this is not a textbook example.

How does the next switch know which VLAN it was?

We add a label: IEEE 802.1Q



inserted here — the only change to the Ethernet frame all semester

12 bits of VLAN ID

4094 usable VLANs

0 and 4095 are reserved

Maximum frame size

1518 → **1522** bytes

the payload floor and ceiling are untouched

Two kinds of link

Trunk — switch to switch, many VLANs, frames tagged

Access — to your laptop, one VLAN, tag removed

Why strip the tag at the access port? Your laptop has no idea what a VLAN is, and the tag would make the frame unreadable to it. The network keeps the complexity to itself — which is why you have never seen one.

We are now at the last layer-2 idea of the course. Everything after the break in Class 4 is physical.

Now cut the cable

Wi-Fi cannot detect collisions — for two separate reasons, and you need both



A radio cannot listen while it shouts

Your own transmission is roughly a million times stronger than the signal you are trying to hear.

It is like trying to hear a whisper across the room while shouting. Full-duplex radio exists in research labs; it is not in your phone.



The collision is not where you are

A collision happens at the RECEIVER, not at the sender.

Two stations far apart may be completely unable to hear each other, yet both reach the access point — where their signals destroy each other. No amount of listening at the sender would reveal it.

We planted the second one in Block 1, on purpose.

So if you cannot detect collisions, you must avoid them instead. CSMA/CA — collision avoidance.

Everything in Block 1 that depended on hearing the crash is now gone. What survives is carrier sensing and backoff.

CSMA/CA in four rules

Round 4 of your Talking Game, adjusted for not being able to hear a crash

1

Sense, then send it all

Channel idle for a short fixed period? Transmit the whole frame. There is no aborting half way — you would not know to abort.

2

Freeze while others speak

Channel busy? Pick a random backoff count and count down only while the channel is idle. Someone starts — freeze. They stop — resume.

3

Zero means go

When the counter reaches zero, transmit. Because the counter froze rather than restarted, whoever waited longest goes first.

4

The receiver must confirm

Every unicast frame is acknowledged. No ACK means the frame was lost, so retransmit with a doubled backoff window.

ONE TRANSMISSION, DRAWN IN TIME



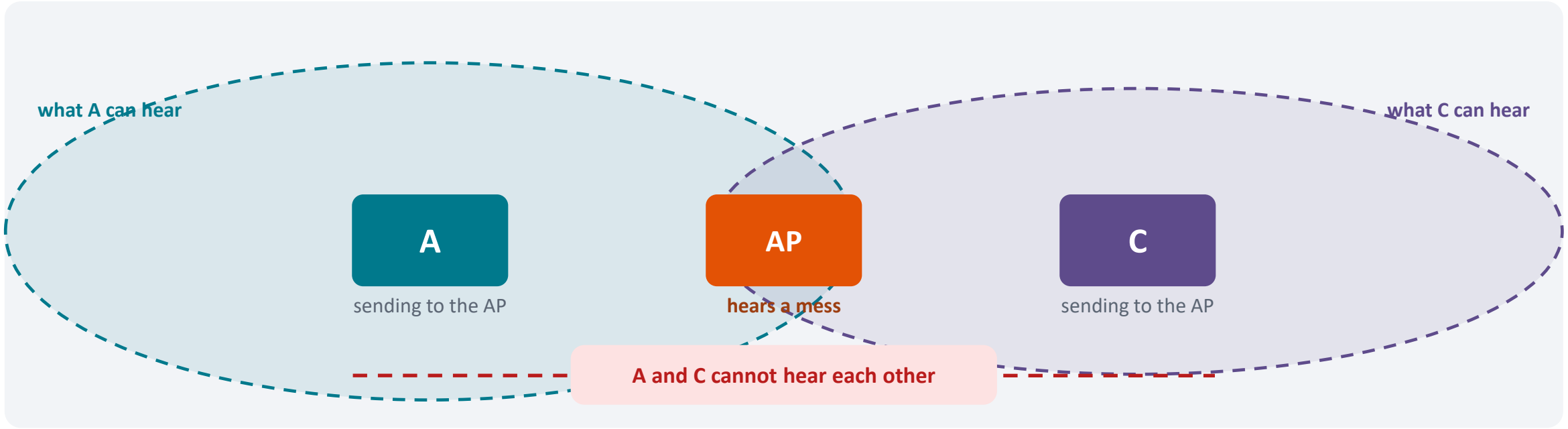
The freeze is the important difference from Ethernet — it is what makes the waiting fair.

This is why Wi-Fi gives you the reliable delivery Ethernet does not. Not because wireless engineers are more careful — because wireless loses frames often enough that waiting for TCP would be far too slow. The cost is airtime, which is why real throughput is far below the number on the box.

DEMONSTRATION · 3 MINUTES

The hidden terminal problem

Two of you send me a private chat at the same moment — nobody else can see them.

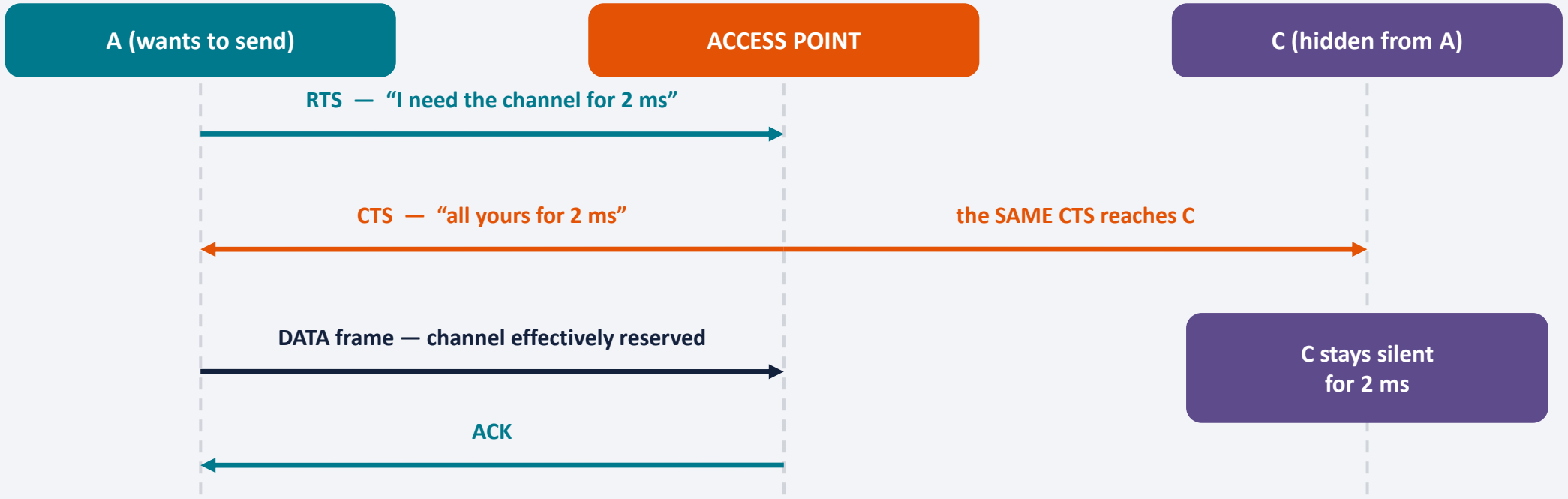


Carrier sensing did not merely fail to help — it actively misled both stations. A listened, heard nothing, and was right: nothing was audible at A. C did the same. Both transmitted. Both were destroyed at the access point.

A concrete wall does the same thing. It is also why Wi-Fi in a large hall behaves far worse than the number of users suggests — stations at opposite ends cannot coordinate at all.

Ask the one station everybody can hear

RTS / CTS — request to send, clear to send



The access point is the one node everybody can hear, so it acts as the referee. That is the whole trick.

The cost is two extra frames per transmission, which is a lot for small frames — so RTS/CTS is usually switched on only above a size threshold, and many networks leave it off. And which family is this? Random access to get the RTS through, then a reservation. Real protocols mix all three.

ONE MORE WI-FI QUESTION

You opened your laptop at home. Your phone was already online.

Nobody typed a password. Nobody plugged anything in. So what happened in those first few seconds?

Who is out there?

beacons, channels and scanning

Which one do I join?

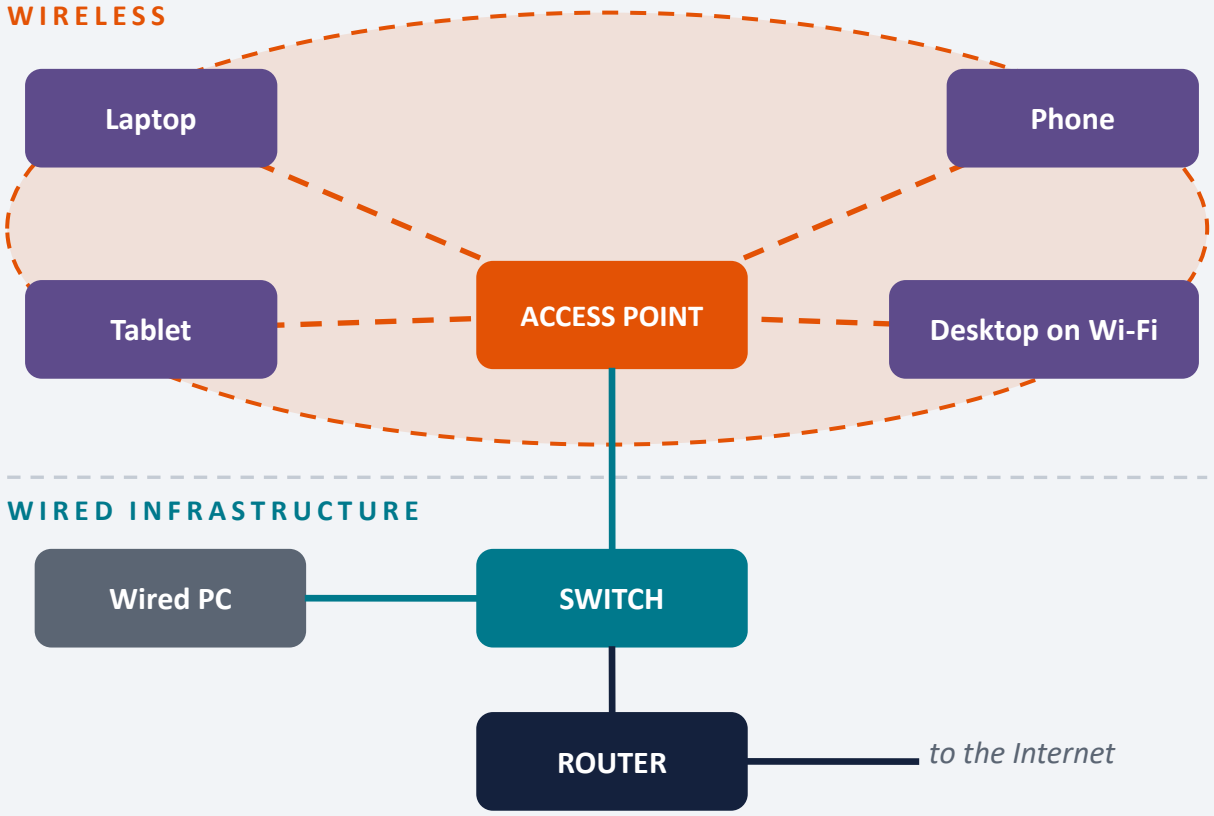
BSS, ESS and association

How do my frames reach the wire?

the 802.11 frame and its three addresses

The pieces of a Wi-Fi network

Kurose names four elements — you already own three of them



Wireless hosts

Laptops, phones. Wireless does not mean mobile — a desktop on Wi-Fi never moves.

Base station = access point

Relays frames between the air and the wired network. A cell tower is a base station too.

Wireless link

The air, plus the multiple access protocol that decides who speaks — CSMA/CA, just now.

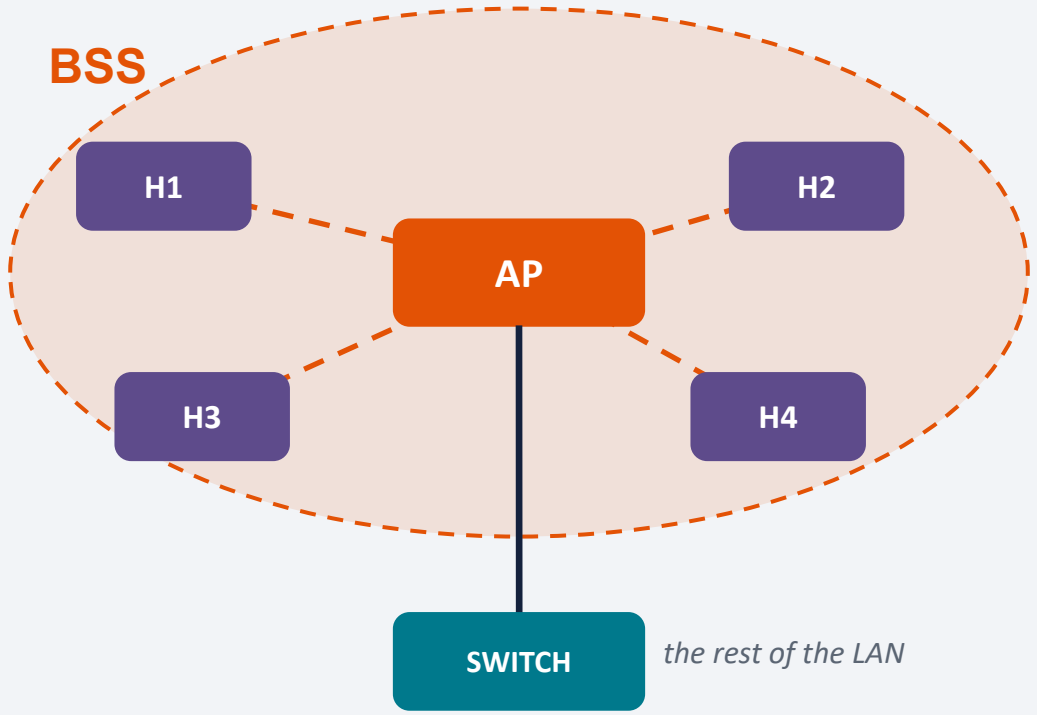
Network infrastructure

The ordinary wired LAN behind the AP: switches, routers, VLANs — all of Block 2.

Keep your eye on the access point. It sits exactly on the border between the two worlds, and this whole section is about what it does there.

One access point and its stations: a BSS

Basic Service Set — Kurose also calls it a cell



The AP is the centre of the cell

In ordinary Wi-Fi every frame goes through the AP — even between two phones a metre apart.

The AP is a bridge, not a router

It takes 802.11 frames off the air and puts 802.3 frames on the wire. It never reads the IP packet inside.

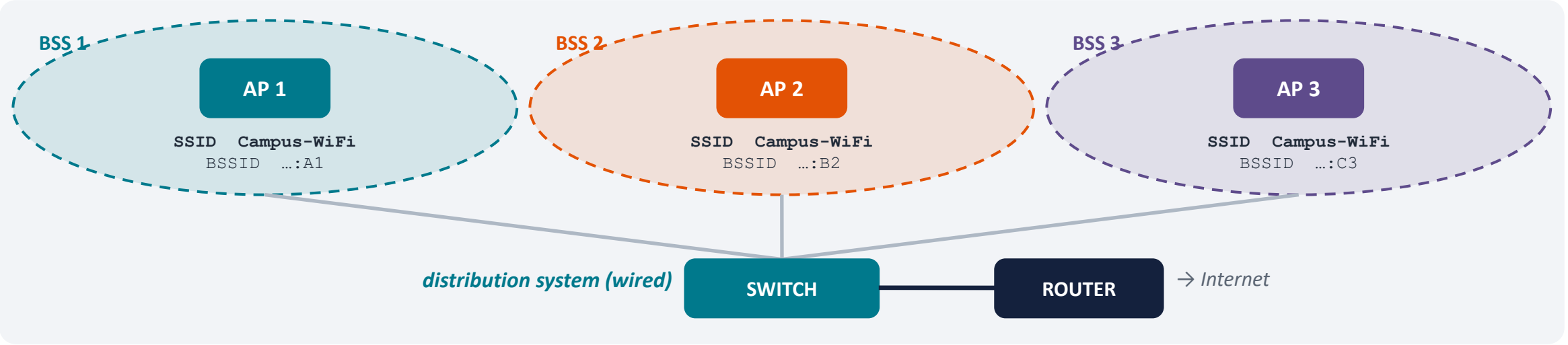
Every BSS has a name and an ID

SSID = the network name on your phone. BSSID = the MAC address of the AP's radio.

Ask: two phones in the same house send a photo to each other. How many times does each frame cross the air?
Twice — phone → AP, then AP → phone.

Many access points, one network: an ESS

Extended Service Set — how a whole building shares one Wi-Fi name



Same SSID everywhere

Your phone shows one network name, however many APs are broadcasting it.

Different BSSID per AP

Your phone always knows which radio it is really talking to.

Joined by a wired distribution system

In practice, the switched Ethernet from Block 2. To the switch, each AP is just a host on a port.

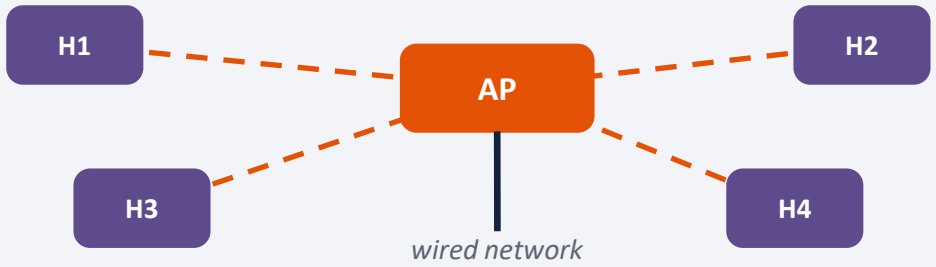
A campus building — or a house with a mesh router — is an ESS. Walk from your room to the kitchen and your phone moves from BSS 1 to BSS 2 without you noticing. That is roaming — we come back to it at the end, and you already know how it works.

Infrastructure mode or ad hoc mode?

Is there an access point — and a wired network behind it?

Infrastructure mode

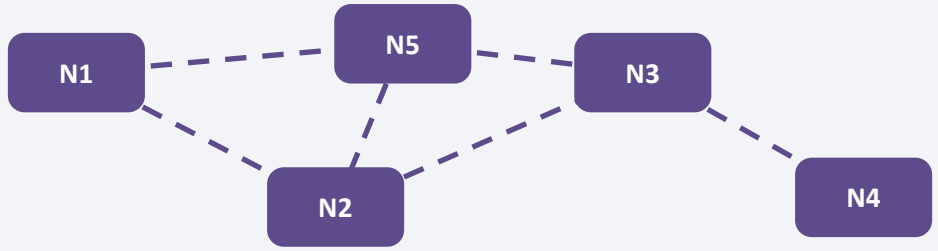
BSS / ESS



- Stations talk only to the AP; the AP relays every frame.
- The AP connects the cell to the wired network and the Internet.
- Moving between APs = handoff (roaming).
- Almost every Wi-Fi network you have ever used.

Ad hoc mode

IBSS — Independent BSS



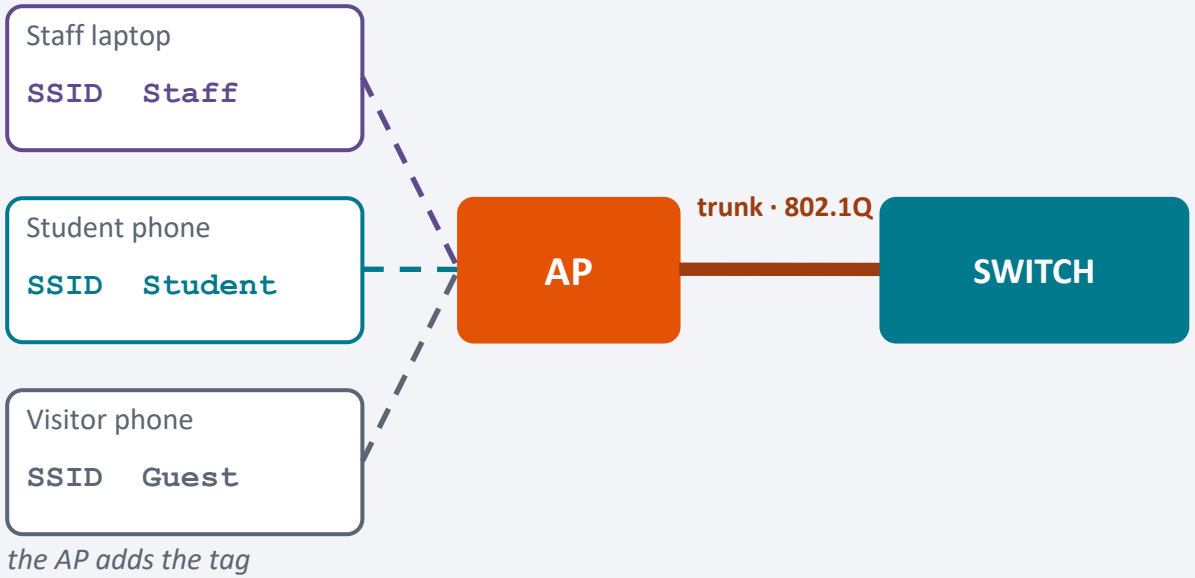
N1 → N4: forwarded by N2 or N5 and N3

- No AP, no wired network — stations talk directly.
- A node reaches only nodes within its radio range.
- To go further, nodes must forward for each other: every node becomes a small router.
- Disaster zones, military units, vehicles, drone swarms.

Common mistake: your phone's personal hotspot is NOT ad hoc mode. The phone turns itself into an access point, and your laptop joins it in infrastructure mode.

Where does Wi-Fi land on the wire? In a VLAN.

One access point, several SSIDs — and each SSID picks a VLAN



1 **One radio, several networks**

Each SSID gets its own BSSID and its own beacons. To your phone they look like three separate networks.

2 **The SSID picks the VLAN**

Every frame from the Student SSID is tagged VLAN 20 before it goes up the trunk to the switch.

3 **Same air, different broadcast domains**

A visitor's phone and a staff laptop on the same AP cannot reach each other at Layer 2 at all.

This is how a campus keeps visitors away from the staff network without building a second Wi-Fi: same access points, same air, separate VLANs behind them. (SSID names and VLAN numbers here are examples.)

First, the AP needs a channel

2.4 GHz is cut into overlapping channels — only three of them do not overlap



Channel centres are 5 MHz apart, but each signal is about 22 MHz wide.

The administrator chooses the channel

Kurose: 11 channels (US rules). Thailand allows 1–13 — the safe set is still 1, 6, 11.

Neighbours on the SAME channel share it

Their stations run CSMA/CA across both networks, so each network gets less.

Neighbours on OVERLAPPING channels are worse

Channel 3 next to channel 6 cannot even take turns — each is just noise to the other.

5 GHz has many more clean channels

A big part of why campuses push devices onto 5 GHz.

Both Block 1 families in one Wi-Fi network: FDM — partitioning by frequency — separates neighbouring networks, and CSMA/CA shares the air inside one channel.

Beacons: every AP keeps announcing itself

About ten times a second, whether or not anyone is listening

Inside one beacon frame

what your phone learns without asking

SSID	`Campus-WiFi`	*the name you see*
BSSID	`…:A1`	*the AP radio's MAC*
Channel	`6`	*where to find it*
Data rates	`1 … 600 Mbps`	*what it can do*
Security	`WPA2 / WPA3`	*how to prove who you are*
Interval	`102.4 ms`	*when the next one comes*

Why so often?

A phone that just walked in must find the network fast — and a moving phone must keep tracking which AP is strongest.

It costs airtime

Beacons go out at a low data rate so every device can decode them. Many APs × several SSIDs = a real share of the air spent on announcements.

"Hidden" SSID?

The AP can leave the name out. Phones then ask for it by name — which broadcasts it anyway. It hides nothing from Wireshark.

Listening for beacons is what Kurose calls passive scanning. Your phone is doing it right now — let's look.

ACTIVITY 3 · 4 MINUTES · ANSWER IN CHAT

Your phone is scanning right now

Open Wi-Fi settings wherever you are. Do not connect to anything new — just look.

1 Count

How many networks does your phone list right now? Type the number in chat — we will find the highest and the lowest.

2 One name, several radios

Does a name appear once although it reaches every room — a mesh router, or the campus SSID? That is an ESS.

3 Strongest signal

Which network has the most bars? Is it the one you are connected to? If not — why might your phone stay put?

4 Compare in chat

Why does one of you see thirty networks and another only three? Give two reasons.

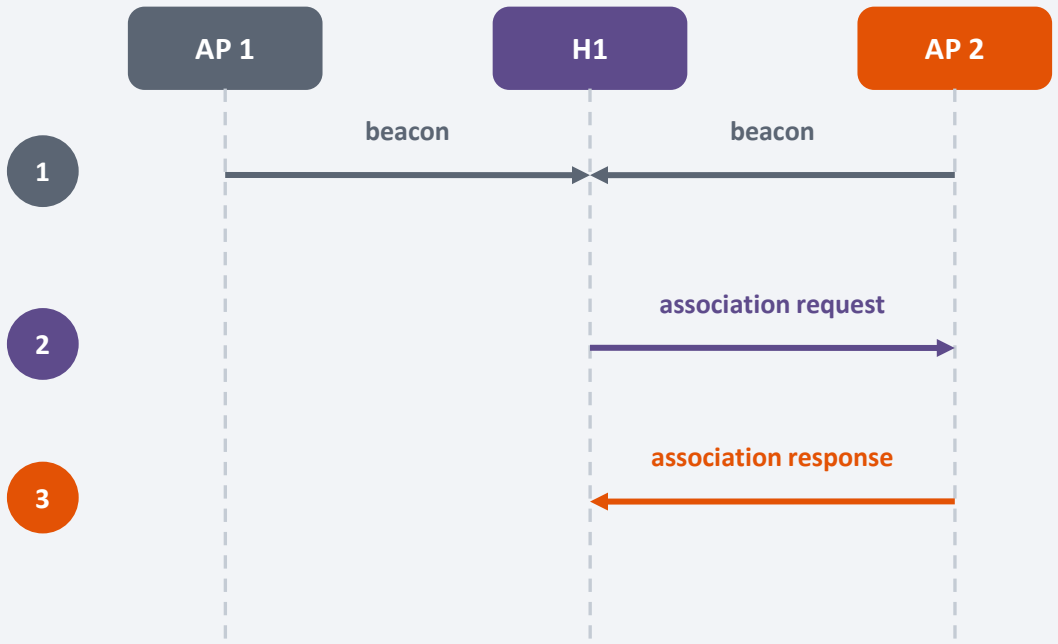
Everything on that list arrived as beacons or probe responses. You typed nothing. A single name covering several access points is an ESS in action.

Tip: some Android phones show the BSSID in the network details; a free Wi-Fi analyser app shows every BSSID and channel. On campus, look for the campus SSID.

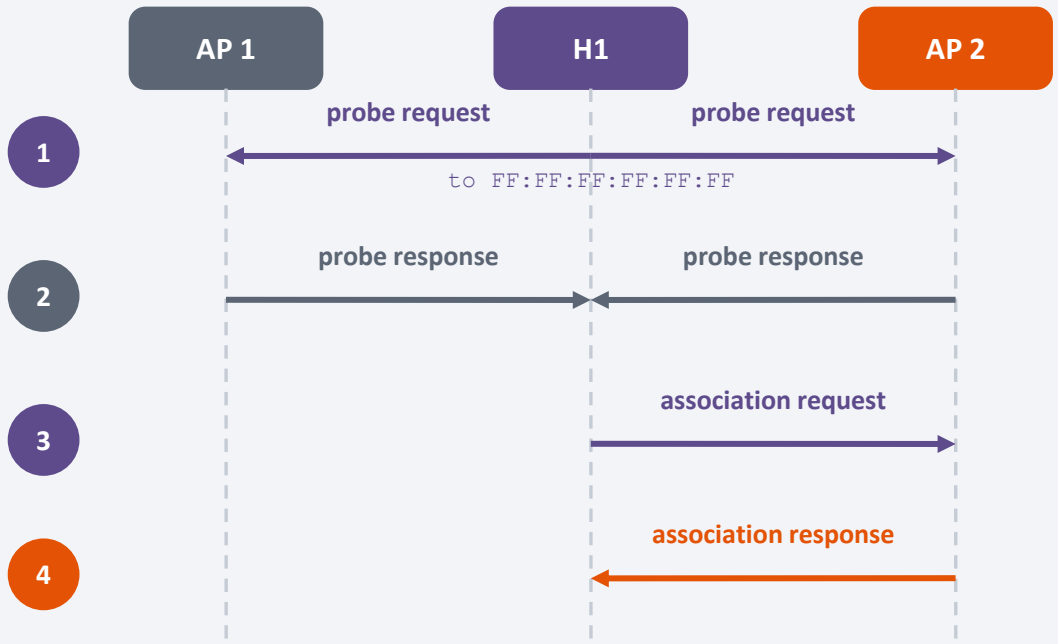
Passive and active scanning

Two ways to find out who is out there — Kurose's figure, step by step

Passive scanning — just listen



Active scanning — ask out loud



Passive is quiet but slow — wait for a beacon on every channel. **Active is fast but noisy** — and a probe can name networks your phone remembers. One more reason phones randomise their MAC while scanning (Class 1).

From “no Wi-Fi” to “online” in six steps

What happened in the first seconds after you walked in

1 Scan

Listen for beacons, or broadcast a probe request.

802.11

2 Choose

Usually a network the phone already knows, and its strongest AP.

your device

3 Associate

Association request and response. The AP now knows your MAC and will relay your frames.

802.11

4 Prove who you are

WPA2 / WPA3 key exchange — password or university account. Kurose ch. 8.

802.11 security

5 DHCP

Get an IP address in the AP’s subnet. The DISCOVER broadcast is simply relayed onto the wire.

Class 11

6 ARP for the gateway

Learn the router’s MAC. Now you are online.

Class 11

Steps 1–4 are 802.11. Steps 5–6 are exactly what happens on any Ethernet. The access point made the air look like an ordinary LAN — which is precisely its job.

Puzzle: why are two addresses not enough?

H1 sends an IP packet to the Internet through router R1. Ethernet has a destination and a source. Which destination?



Option A `destination = AP` ✗

The AP catches it — good. But the AP is a bridge; it does not read IP. Once it has the frame, nothing in it names anyone on the wire. Where should it go next?

Option B `destination = R1` ✗

Names the right final device — good. But R1 is not on the air. Which radio should pick this out of the air, relay it, and send the ACK? The AP has no reason to think it is for it.

Neither works alone. The frame must say who catches it on the air AND where it is really going on the LAN — plus who sent it. **That is three addresses.**

Vote before you click on: A, B, or neither?

The 802.11 frame

Kurose’s figure, every field named — sizes in bytes



COMPARE: THE ETHERNET FRAME



← *two addresses, no reason for more on a cable*

Same CRC-32 as Ethernet

Exactly the error detection from the Class 2 video.

A bigger payload

Up to 2312 bytes, against Ethernet’s 1500.

No preamble in the frame

On a radio, synchronisation is done by the physical layer — Class 5.

Next: what the three addresses mean — then Duration and Seq control, which solve problems you met earlier today.

Which address is which?

Answer three questions and you can fill in any 802.11 frame

Address 1

Who should catch this on the air?

The receiving radio on this hop — an AP or a station.

Address 2

Who put it on the air?

The transmitting radio on this hop — and who gets the ACK.

Address 3

Who is at the other end on the wired side?

Going up: the final destination on the LAN. Coming down: the original source.

Address 4

Only when two APs forward to each other over the air

Mesh networks and wireless bridges. Ignore it for ordinary Wi-Fi.

Direction	Addr 1	Addr 2	Addr 3
station → AP (towards the wire)	AP	station	LAN destination
AP → station (from the wire)	station	AP	LAN source

The pattern to remember

Addresses 1 and 2 are always the two radios on this hop.

Address 3 is always the device on the wired side.

The AP's own MAC appears only while the frame is in the air.

Kurose's slide says Address 4 is "used only in ad hoc mode". The standard uses it for AP-to-AP forwarding.

Worked example: H1 sends to the Internet

Kurose's example — follow one frame from the air onto the wire



On the air — 802.11 frame

H1 → AP

Address 1	AP	*receiver on the air*
Address 2	H1	*transmitter on the air*
Address 3	R1	*where it is really going*

AP converts

On the wire — 802.3 frame

AP → switch → R1

Destination	R1	*copied from Address 3*
Source	H1	***still H1 — NOT the AP!***
Payload	IP pkt	*untouched*

The AP's own address has vanished. On the wire the frame looks as if H1 were plugged straight into the switch — so the switch learns "H1 is behind the AP's port", and the reply can find its way back.

The AP is as invisible to the wired network as a switch is to its hosts.

And the reply: R1 sends back to H1

The same rule, in the other direction



On the wire — 802.3 frame

R1 → switch → AP

Destination	`H1`	*the switch knows: AP's port*
Source	`R1`	*the router*
Payload	`IP pkt`	*untouched*

AP converts

On the air — 802.11 frame

AP → H1

Address 1	`H1`	*receiver on the air*
Address 2	`AP`	***now the transmitter***
Address 3	`R1`	*original source, wired side*

Addresses 1 and 2 simply swapped roles; Address 3 is still R1. The two radios on this hop, plus the device on the wired side — in both directions.

How did the switch know which port leads to H1? It learned it from the previous frame — the Human Switch, again.

Three more fields you already understand

Each one exists because of a problem you met earlier today

Frame control — 2 bytes

Type / subtype

data, ACK, RTS, CTS, beacon, probe, association…

To DS / From DS

direction relative to the wire — how to read Addr 1–3

Retry

“this is a retransmission”

Earlier: every frame type in this section

Duration — 2 bytes

What it says

how long I need the channel

Who reads it

every station that hears the frame

What they do

set a timer and stay quiet until it expires

Earlier: RTS/CTS — this is how C knew to stay silent for 2 ms

Sequence control — 2 bytes

What it says

a number for every frame

The problem

the ACK is lost, so the sender retransmits

The fix

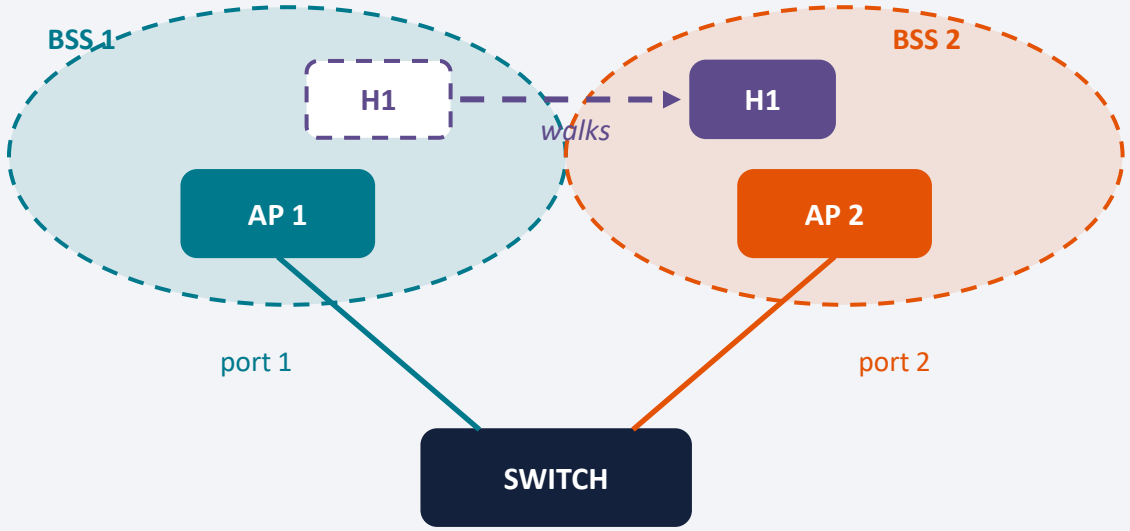
same number twice → receiver drops the duplicate

Earlier: CSMA/CA rule 4 — now safe to retransmit

Every one of these fields exists because of a problem the air creates. Ethernet has none of those problems — so Ethernet has none of these fields.

Walking down the corridor: roaming

Kurose: mobility within the same subnet



1. **Signal fades**
   AP 1 gets weaker as H1 walks; AP 2 gets stronger.

2. **H1 reassociates with AP 2**
   Same SSID, different BSSID — still the same ESS.

3. **The IP address stays the same**
   Both APs are on the same subnet. Nothing for DHCP to change.

4. **The switch re-learns the port**
   The first frame from H1 arrives on port 2 — the learn rule updates the table. The Human Switch, again.

Nothing new was invented: roaming inside one subnet is switch self-learning. Moving to a different subnet would change the IP address — a Layer 3 problem, beyond this course.

QUICK CHECK · 3 MIN

Joining a Wi-Fi network

Q1 Your phone’s personal hotspot runs in…

- A ad hoc mode
- B infrastructure mode
- C IBSS mode
- D mesh mode

Q2 Passive scanning means the phone…

- A sends probe requests
- B listens for beacons
- C asks the DHCP server
- D pings every AP

Q3 H1 → AP → router R1. What is Address 3?

- A the AP’s MAC
- B H1’s MAC
- C R1’s MAC
- D FF:FF:FF:FF:FF:FF

Q4 You roam to another AP in the same subnet. What changes?

- A your IP address
- B the switch’s table
- C the SSID
- D your MAC address

Wi-Fi networks in six lines

How your phone got online, and what its frames look like

1	**BSS, ESS, IBSS**	One AP and its stations · many APs, one SSID, wired together · no AP at all (ad hoc).
2	**Beacons**	Every AP announces its SSID and BSSID about ten times a second.
3	**Scanning**	Passive: listen for beacons. Active: broadcast a probe request and wait for responses.
4	**Joining**	Scan → associate → prove who you are → then DHCP and ARP, as on any LAN.
5	**Three addresses**	Receiver on the air, transmitter on the air, and the device on the wired side.
6	**Roaming**	Reassociate, keep the IP address — and the switch simply re-learns your port.

Ethernet and Wi-Fi, side by side

The exam will ask you to compare them

	ETHERNET (switched) 802.3	WI-FI 802.11
Frame idea, MAC addresses, CRC	the same	the same
Contention method	CSMA/CD — detect and abort	CSMA/CA — avoid, never abort
Acknowledgments	none at all	every unicast frame
Medium	a dedicated port per device	shared air, everyone at once
Duplex	full — send and receive together	half — one at a time
Can it see the collision?	yes, at the sender	no — it happens at the receiver
Collisions in practice	essentially impossible today	normal, constant, designed around

Six things to leave with

If you can say all six in your own words, today worked

1	**Three solutions only**	Sharing a channel offers exactly three options: partition it, let everyone try, or take turns.
2	**Sensing is not enough**	Carrier sensing helps but cannot eliminate collisions — propagation delay guarantees it.
3	**64 bytes is physics**	The minimum Ethernet frame is the time a collision takes to travel back to the sender.
4	**A switch teaches itself**	It learns from source addresses, forwards to one port, and floods when it does not know.
5	**Who splits what**	Switches split collision domains. Routers split broadcast domains. VLANs split broadcast domains in software.
6	**Wi-Fi avoids and confirms**	It cannot detect collisions, so it avoids them and acknowledges every single frame.

THAT IS THE END OF LAYER 2

Next: down to the wire and the air themselves

Class 4 · recorded video

- Transmission media, signals and line coding: how a 1 and a 0 actually travel down a wire or through the air.
- Watch it before we meet again — Class 5 assumes it.
- There will be a warm-up quiz on it, exactly like today's.

Class 5 · LIVE in the room

- Bandwidth, Shannon capacity, modulation and multiplexing, plus the course wrap-up.
- Cables passed around: twisted pair, coax, single-mode and multimode fibre, connectors, crimping tool.
- Data across the room by phone flashlight.

To think about while you watch Class 4: your Wi-Fi says 1200 Mbps on the box. Why do you never see it? You already know part of the answer from today. The rest is Layer 1.